

WEEK 6 ASSIGNMENT

Git, GitHub & Final Data Analytics Project

Data Analytics Internship

Project Title:

E-Commerce Sales Data Analytics

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Course: BCA – Final Year

Internship Program: InternNova

Domain: Data Analytics

Assignment Duration: 1 Week (7 Days)

Total Marks: 100

Submission Format: GOOGLE DRIVE LINK UPLOAD

TASK 1: Git & GitHub Basics

Aim

To learn and implement the basic Git and GitHub workflow by creating a repository, cloning it to the local system, adding project files, committing changes, and pushing them to GitHub.

Tools Used

- Git
- GitHub
- Command Prompt (CMD)
- Windows File Explorer

Introduction

Git is a version control system used to track changes in project files. GitHub is an online platform used to store and manage Git repositories. In this task, Git and GitHub were used to create and manage the E-Commerce Sales Data Analytics project repository.

Steps Performed

1. Git Installation Verification

Git was installed successfully on the system and its version was checked using the following command:

```
git --version
```

```
Microsoft Windows [Version 10.0.26200.9168]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Anjali>git --version  
git version 2.55.0.windows.5
```

2. GitHub Repository Creation

A GitHub repository named **ecommerce-sales-data-analytics** was created for the project.

Repository URL:

<https://github.com/Mayur320/ecommerce-sales-data-analytics>

3. Clone GitHub Repository

The repository was cloned from GitHub to the local computer using:

```
git clone https://github.com/Mayur320/ecommerce-sales-data-analytics.git
```

```
C:\Users\Anjali\Documents>git clone https://github.com/Mayur320/ecommerce-sales-data-analytics.git
Cloning into 'ecommerce-sales-data-analytics'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

4. Add Project Dataset

The E-Commerce Sales dataset was added to the local repository inside the Dataset folder.

5. Configure Git User

Git username and email were configured using:

```
git config --global user.name "Mayur"
git config --global user.email "your-github-email"
```

6. Add Files to Staging Area

The project files were added to the Git staging area using:

```
git add .
```

7. Commit Changes

The dataset was committed with a meaningful commit message:

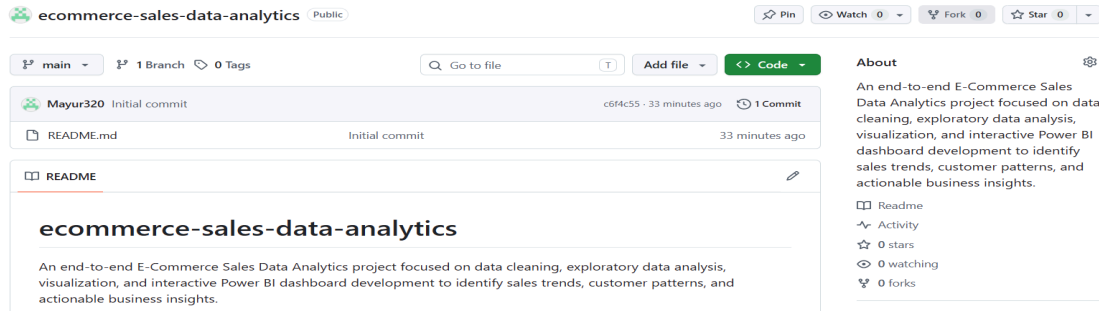
```
git commit -m "Add e-commerce sales dataset"
```

```
C:\Users\Anjali\Documents\ecommerce-sales-data-analytics>git commit -m "Add e-commerce sales dataset"
[main f143960] Add e-commerce sales dataset
1 file changed, 253 insertions(+)
create mode 100644 1. Dataset/ecommerce_sales_dataset.csv
```

8. Push Files to GitHub

The committed files were uploaded to the GitHub repository using:

`git push`



Result

The GitHub repository was successfully created, cloned, and configured. The E-Commerce Sales dataset was successfully committed and pushed to the GitHub repository.

Conclusion

This task provided practical knowledge of Git and GitHub. The basic process of creating a repository, cloning it, adding files, committing changes, and pushing files to GitHub was successfully completed.

TASK 2: Git Workflow

Aim

To practice the basic Git workflow for tracking, staging, committing, and synchronizing project files with a GitHub repository.

Git Workflow

The Git workflow followed in this project was:

Working Directory → Staging Area → Local Repository → GitHub Repository

Steps Performed

1. Check Repository Status

The repository status was checked using:

```
git status
```

Initially, the dataset was shown as an untracked file.

2. Add Files to Staging Area

The files were added to the staging area using:

```
git add .
```

3. Verify Staged Files

The repository status was checked again:

```
git status
```

The dataset appeared under:

Changes to be committed

```
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
       new file:   1. Dataset/ecommerce_sales_dataset.csv
```

4. Commit Changes

The staged dataset was committed using:

```
git commit -m "Add e-commerce sales dataset"
```

The commit was successfully created.

```
C:\Users\Anjali\Documents\ecommerce-sales-data-analytics>git commit -m "Add e-commerce sales dataset"
[main f143960] Add e-commerce sales dataset
1 file changed, 253 insertions(+)
create mode 100644 1. Dataset/ecommerce_sales_dataset.csv
```

5. Push Changes to GitHub

The committed changes were pushed to the remote GitHub repository:

```
git push
```

The push was completed successfully with:

```
main -> main
```

```
C:\Users\Anjali\Documents\ecommerce-sales-data-analytics>git push
info: please complete authentication in your browser...
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 5.83 KiB | 5.83 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Mayur320/ecommerce-sales-data-analytics.git
c6f4c55..f143960 main -> main
```

Result

The Git workflow was successfully performed. The E-Commerce Sales dataset was checked, staged, committed, and pushed to the GitHub repository.

Conclusion

The practical Git workflow of **status** → **add** → **commit** → **push** was successfully implemented. This process helps maintain project versions and synchronize local project files with GitHub.

Task 3: Dataset Selection & Data Preparation

Aim

To select an E-Commerce Sales dataset and prepare it for further analysis by cleaning missing values, duplicate records, and date formats.

Dataset Description

The dataset contains e-commerce order information such as:

- Order ID
- Order Date
- Region
- Category
- Customer Segment
- Payment Method
- Units Sold
- Unit Price
- Discount
- Sales
- Profit

Data Cleaning Performed

1. Checked the dataset structure and data types.
2. Checked for missing values.
3. Identified duplicate records.
4. Removed duplicate rows.
5. Replaced missing values in `Customer_Segment` with `Consumer`.
6. Converted `Order_Date` into proper date format.
7. Prepared the cleaned dataset for EDA and visualization.

Result

The dataset was successfully cleaned and prepared for further analysis.

Task 4: Exploratory Data Analysis (EDA)

Aim

To explore the cleaned e-commerce dataset and identify important patterns and relationships in sales and profit.

Analysis Performed

1. Descriptive Statistics

Used `describe()` to understand numerical columns such as:

- Units Sold
- Unit Price
- Discount
- Sales
- Profit

2. Sales by Category

Calculated total sales for each product category to identify the best-performing categories.

3. Sales by Region

Calculated total sales for each region to compare regional performance.

4. Sales by Customer Segment

Analyzed sales generated by different customer segments.

5. Profit by Category

Compared total profit generated by different product categories.

6. Correlation Analysis

Created a correlation matrix for:

- Units Sold
- Unit Price
- Discount
- Sales
- Profit

Result

EDA helped identify important sales, profit, regional, category, and customer-segment patterns in the dataset.

Task 5: Data Visualization

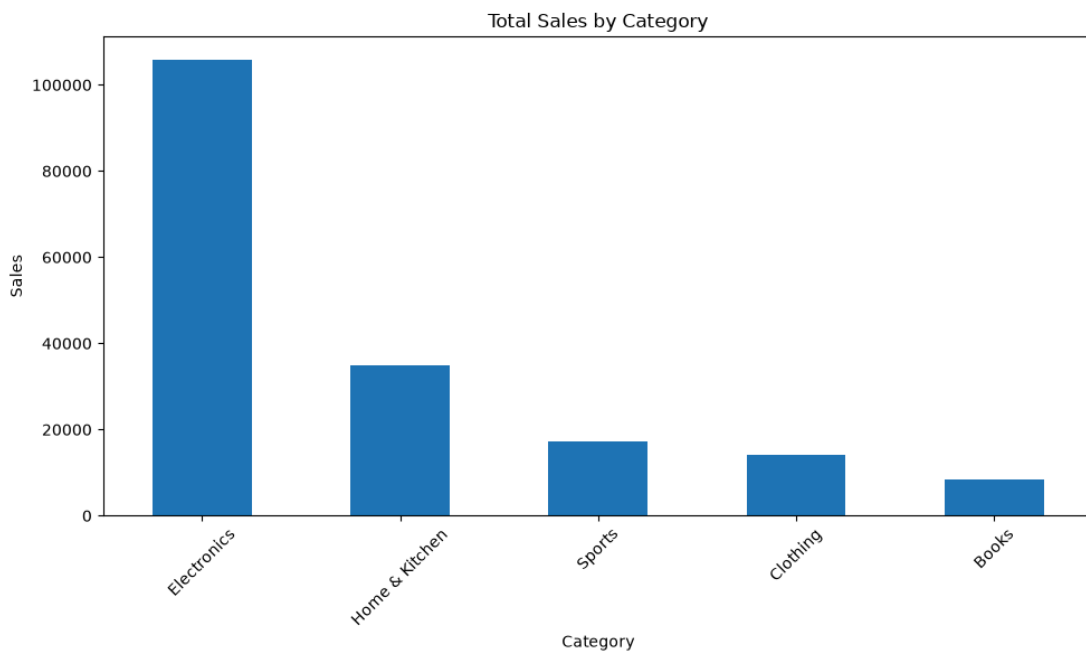
Aim

To create meaningful visualizations using Python Matplotlib for better understanding of the e-commerce data.

Visualizations Created

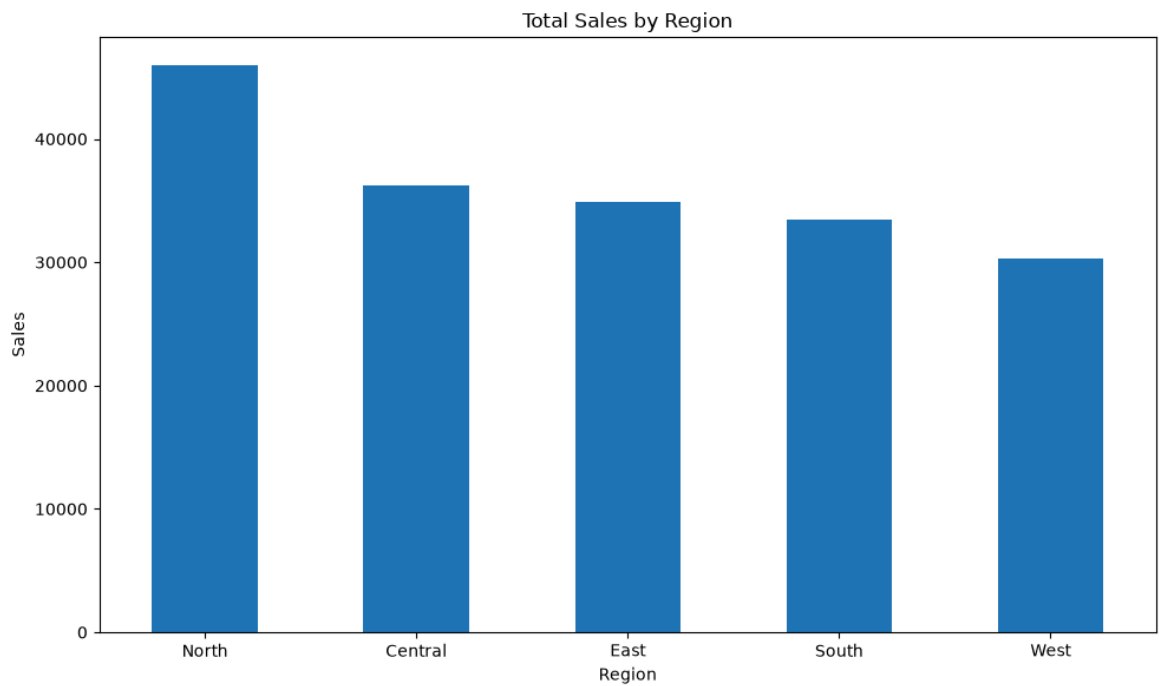
1. Total Sales by Category

A bar chart was created to compare sales across different product categories.



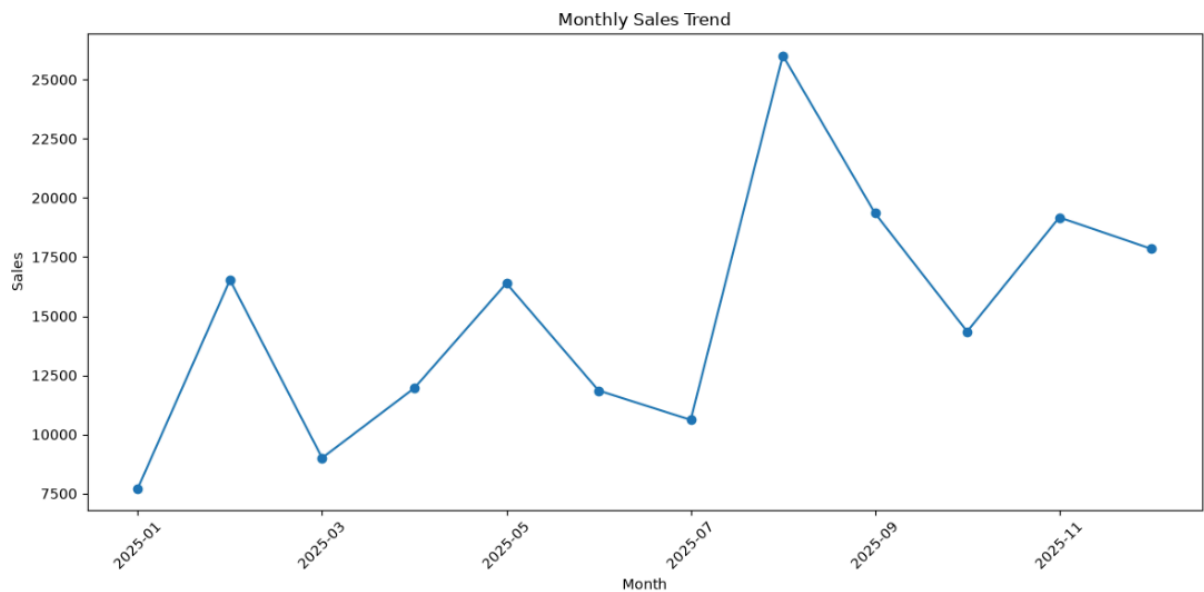
2. Total Sales by Region

A bar chart was created to compare sales performance across regions.



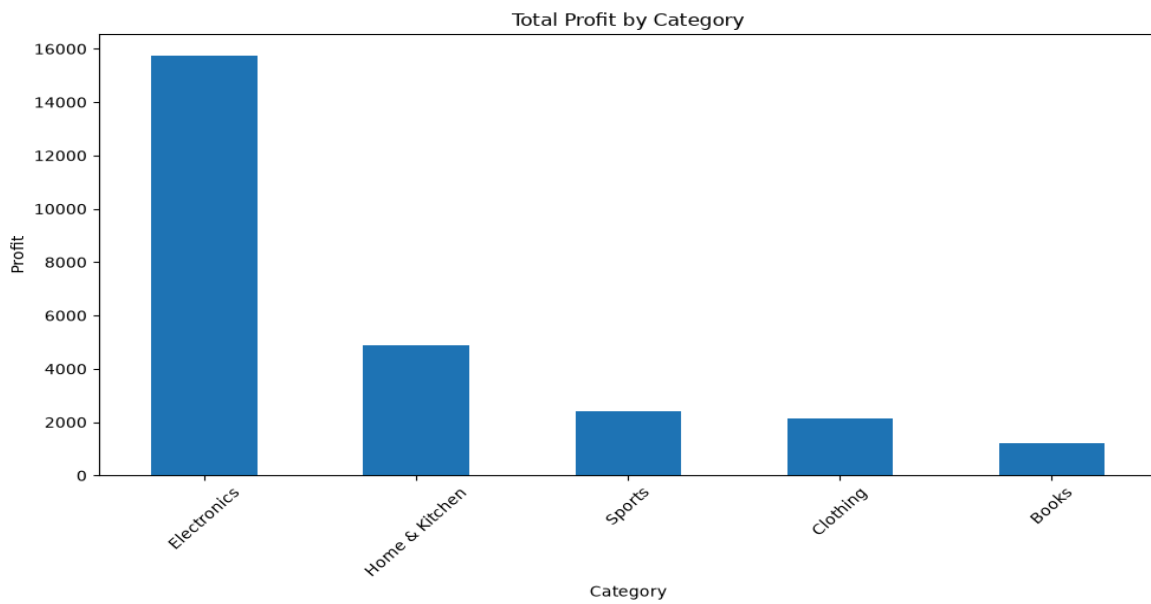
3. Monthly Sales Trend

A line chart was created to understand how sales changed over different months.



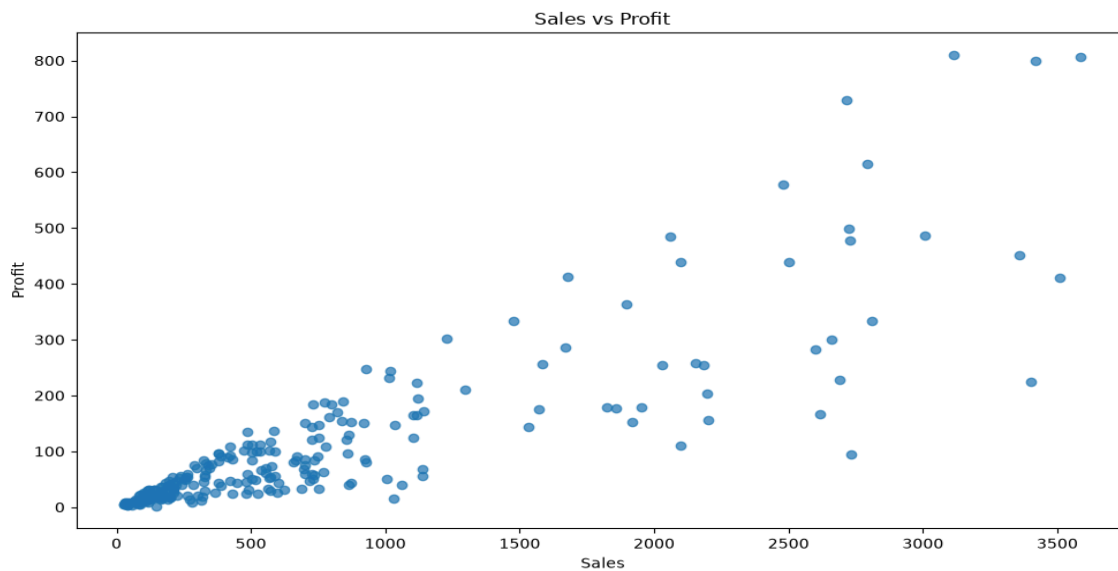
4. Total Profit by Category

A bar chart was created to compare profit generated by each category.



5. Sales vs Profit

A scatter plot was created to study the relationship between sales and profit.



Result

The visualizations made it easier to understand sales trends, category performance, regional performance, and the relationship between sales and profit.

Task 6: Power BI Dashboard Development

Aim

To create an interactive Power BI dashboard for analyzing E-Commerce Sales data using KPI cards, charts, and slicers.

Dashboard Components

The Power BI dashboard was created using the cleaned E-Commerce Sales dataset. The following components were included:

1. **Total Sales KPI** – Displays the overall sales generated.
2. **Total Profit KPI** – Displays the total profit generated.
3. **Total Units Sold KPI** – Shows the total number of units sold.
4. **Sales by Category** – A column chart used to compare sales across product categories.
5. **Sales by Region** – A bar chart used to compare sales across different regions.
6. **Monthly Sales Trend** – A line chart showing sales performance over time.
7. **Profit by Category** – A column chart comparing profit generated by each category.
8. **Sales vs Profit** – A scatter chart used to analyze the relationship between sales and profit.
9. **Region Slicer** – Allows users to filter the dashboard by region.
10. **Customer Segment Slicer** – Allows users to filter the dashboard by customer segment.

Tools Used

- Microsoft Power BI
- Cleaned E-Commerce Sales Dataset
- Power BI Card, Column Chart, Bar Chart, Line Chart, Scatter Chart and Slicer visuals

Explanation

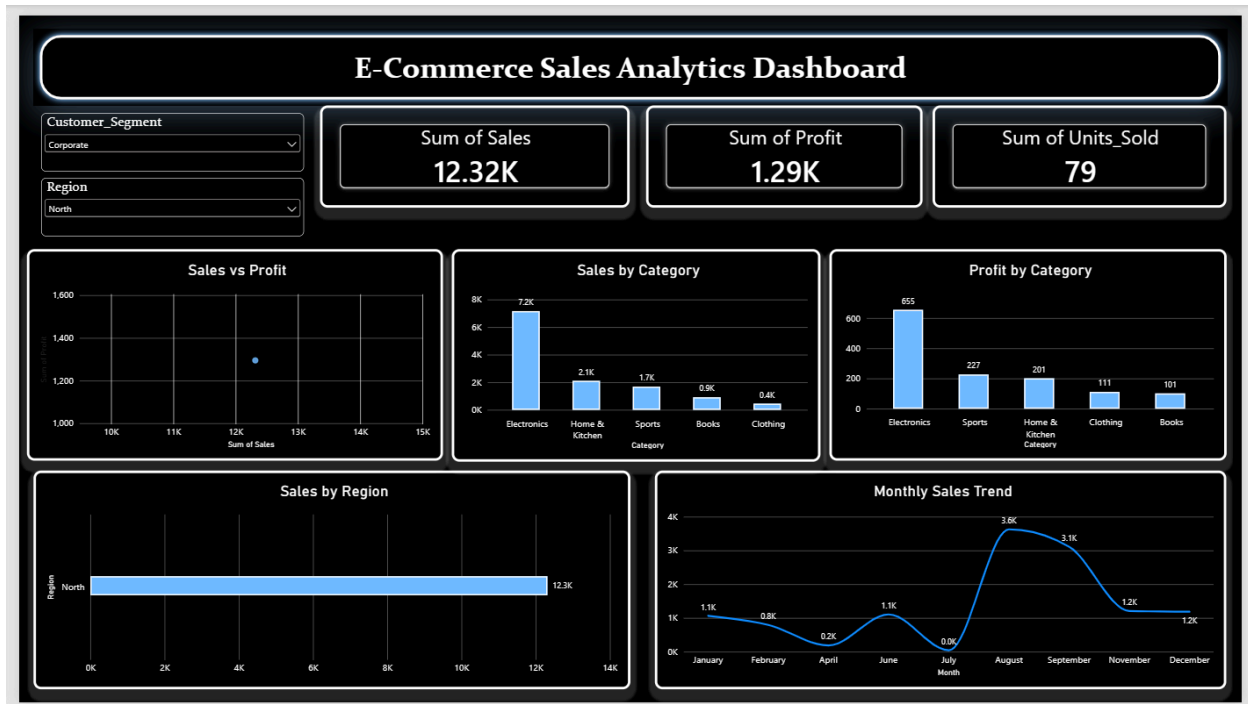
The dashboard provides an interactive view of the E-Commerce sales data. KPI cards provide a quick summary of important business metrics, while charts help analyze category-wise, region-wise, monthly, and profit-related performance. Slicers allow users to interactively filter the dashboard based on region and customer segment.

Expected Outcome

The dashboard helps users quickly understand sales performance, profitability, regional performance, customer segments, and monthly sales trends. It supports better business analysis and decision-making.

Conclusion

An interactive Power BI dashboard was successfully developed with **3 KPI cards**, **5 visualizations**, and **2 slicers**. The dashboard provides a clear and user-friendly representation of E-Commerce sales data and helps identify important business patterns and performance trends.



FINAL DASHBOARD

Task 7: Business Insights & Recommendations

Business Insights

1. Category-wise sales analysis helps identify the best-performing product categories.
2. Region-wise sales analysis shows which regions generate higher sales.
3. Customer segment analysis helps understand the purchasing contribution of different customer groups.
4. Profit analysis helps identify categories contributing more to overall profitability.
5. Monthly sales analysis helps understand changes in sales performance over time.

Recommendations

1. Focus marketing and inventory planning on high-performing product categories.
2. Improve promotional strategies in lower-performing regions and customer segments.

Conclusion

The Jupyter Notebook analysis successfully cleaned and explored the E-Commerce Sales dataset. Different statistical analyses and visualizations were used to understand sales, profit, categories, regions, customer segments, and trends. These findings can be further used to create an interactive **Power BI dashboard** for business decision-making.

Task 8: GitHub Documentation & Project Presentation

Aim

To maintain professional project documentation on GitHub and prepare a short presentation explaining the E-Commerce Sales Data Analytics project.

GitHub Documentation

A GitHub repository was created for the project:

Repository Name: `ecommerce-sales-data-analytics`

The repository contains the following project files and folders:

- Dataset
- Python Notebook
- Visualizations
- Power BI Dashboard
- Project Presentation
- Final Project Report
- README.md

The README file provides information about the project, objectives, tools used, methodology, analysis, dashboard, insights, and conclusion.

Project Presentation

A short presentation was prepared to explain the complete project workflow. The presentation covers:

1. Project Title
2. Problem Statement
3. Objectives
4. Dataset
5. Data Cleaning
6. Exploratory Data Analysis
7. Data Visualizations
8. Power BI Dashboard
9. Business Insights
10. Recommendations
11. Conclusion

Documentation Standards

The project was organized using proper folder names and meaningful file names. Git and GitHub were used to maintain and share the project files and track project development.

Conclusion

The project was professionally documented and organized on GitHub. A project presentation was also prepared to communicate the methodology, analysis, dashboard, findings, and recommendations clearly.

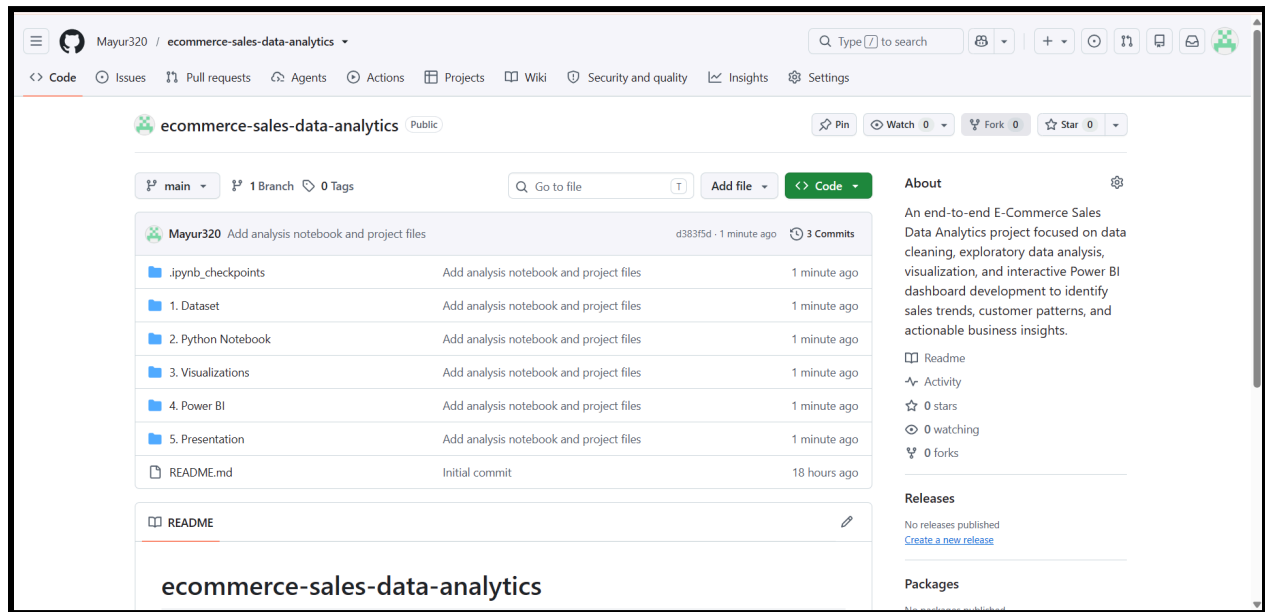


Figure: GitHub Repository for E-Commerce Sales Data Analytics Project